

Introduction

In today's scenario, most real-world systems can be represented as networks that change over time. Examples include a communication network in which interactions occur at specific times, a scheduled transport system, a social network formed by time-constrained interactions, and information on disease-spreading processes. In all these processes, time is an important factor in determining how influence and information entities propagate through the network. Static graph information that ignores temporal information is insufficient for representing the system's dynamics.

Temporal networks extend graph models by associating time with the edges, supporting the study of time-dependent connectivity. The core concept within time-sensitive network analysis is the temporal path constraint, which is a sequence of edges in strictly increasing time order. Temporal paths form the basis of temporal connectivity, which determines when a node can be reached from a given source under temporal constraints. This concept shows to be important to understand dynamic mechanisms, like information diffusion, epidemic spreading, and network resistance in time-evolving systems.

The algorithmic foundations of temporal connectivity are well-studied, including simple and efficient approaches such as temporal breadth-first search (BFS). The existing graph visualisation tools, designed for static networks, provide limited support for exploring the temporal constraints and observing how connectivity evolves over time. This results in a mismatch for conceptual models of temporal networks and easy-to-use tools that allow users to explore them.

The Temporal Path Explorer and Visualizer project aims to handle this gap via designing and implementing a web-based tool to explore temporal paths with changing networks. The system will allow users to upload time-based network data and visually see how connectivity from a selected node source evolves over time. Linking the temporal connectivity algorithm with visualisation, this tool makes the concept of time-constrained paths explicit and concrete, that is enabling users to observe the impact of temporal ordering on network connectivity.

The core of the system will be developed for the implementation of a simple temporal connectivity algorithm, which will compute reachable nodes stepwise, such as a one-pass temporal BFS. The algorithm's results are reflected immediately in the visual interface, facilitating users to follow the temporal expansion of reachable nodes stepwise. This merging of computation and visualisation supports exploratory analysis and helps to connect that links the theoretical and practical understanding.

The distinctive feature of the proposed system is its 'what-if' analysis functionality. Users will be able to remove or delay certain temporal edges and observe how these changes will affect the connectivity throughout the network. This functionality draws attention to the temporal processes to individual events and enables users to experiment with unanticipated scenarios, such as delayed communication or missing connections. This experimentation is useful to understand robustness in temporal networks.

The project is primarily implementation-driven, focusing on developing a functional, interactive software system. It also indulges supporting research elements, including a review of existing work on time-varying networks, temporal path algorithms, data structures, and the visual encoding used in the system. The project adopts a approach that reflects most recent software development practices, with implementation guided by research and work.

To manage the complexity of developing an algorithm-driven visual analytics tool, the project follows a structured software engineering approach. This includes clear requirements analysis, careful system design, and the adoption of an appropriate software development life cycle. The project is carried out collaboratively with a team, reflecting professional practice in which effective communication, task coordination, and documentation are essential for successful delivery.

The temporal path explorer and visualizer project aims to provide an intuitive and educational interface for exploring temporal connectivity among dynamic networks, while simultaneously giving students practical experience in software engineering, algorithm implementation, and interactive visualisation. The work undertaken in this project also serves as valuable preparation for the MSc dissertation, where equivalent skills, planning, execution, and detailed evaluation are required.

References

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